

MTH-262 Statistics and Probability Theory

Lecture 02



Outlines of Today's Lecture

- In this lecture we will discuss following topics:
- Data Presentation: Frequency distribution, Graphical Presentation.
- Measures of Central Tendency: Arithmetic Mean.

Data Presentation

- Frequency Distribution

Presenting number of observations according to their class is known as the frequency distribution.

Example (Revisited).

Table 1.4: Car Battery Life

2.2	4.1	3.5	4.5	3.2	3.7	3.0	2.6
3.4	1.6	3.1	3.3	3.8	3.1	4.7	3.7
2.5	4.3	3.4	3.6	2.9	3.3	3.9	3.1
3.3	3.1	3.7	4.4	3.2	4.1	1.9	3.4
4.7	3.8	3.2	2.6	3.9	3.0	4.2	3.5

How to Construct a Frequency Distribution

- 1: Sort the data in ascending order.
- 2: Calculate the range of data.
- 3: Decide on the number of intervals in the **frequency distribution**.
- 4: Determine the intervals.
- 5: Tally and count the observations under each interval.

$$\text{Range} = X_m - X_0$$

$$\text{No. of Classes} = \text{Range}/h$$

$$h = \text{Class Interval}$$

Table 1.7: Relative Frequency Distribution of Battery Life

Class Interval	Class Midpoint	Frequency, f	Relative Frequency
1.5–1.9	1.7	2	0.050
2.0–2.4	2.2	1	0.025
2.5–2.9	2.7	4	0.100
3.0–3.4	3.2	15	0.375
3.5–3.9	3.7	10	0.250
4.0–4.4	4.2	5	0.125
4.5–4.9	4.7	3	0.075

Data Presentation

- Simple Bar Chart

A simple bar chart consists of horizontal or vertical bars of equal width and length proportional to the values they present. As the basis of comparison is linear or one dimensional, the width of these bars have no significance but to make the chart look attractive.

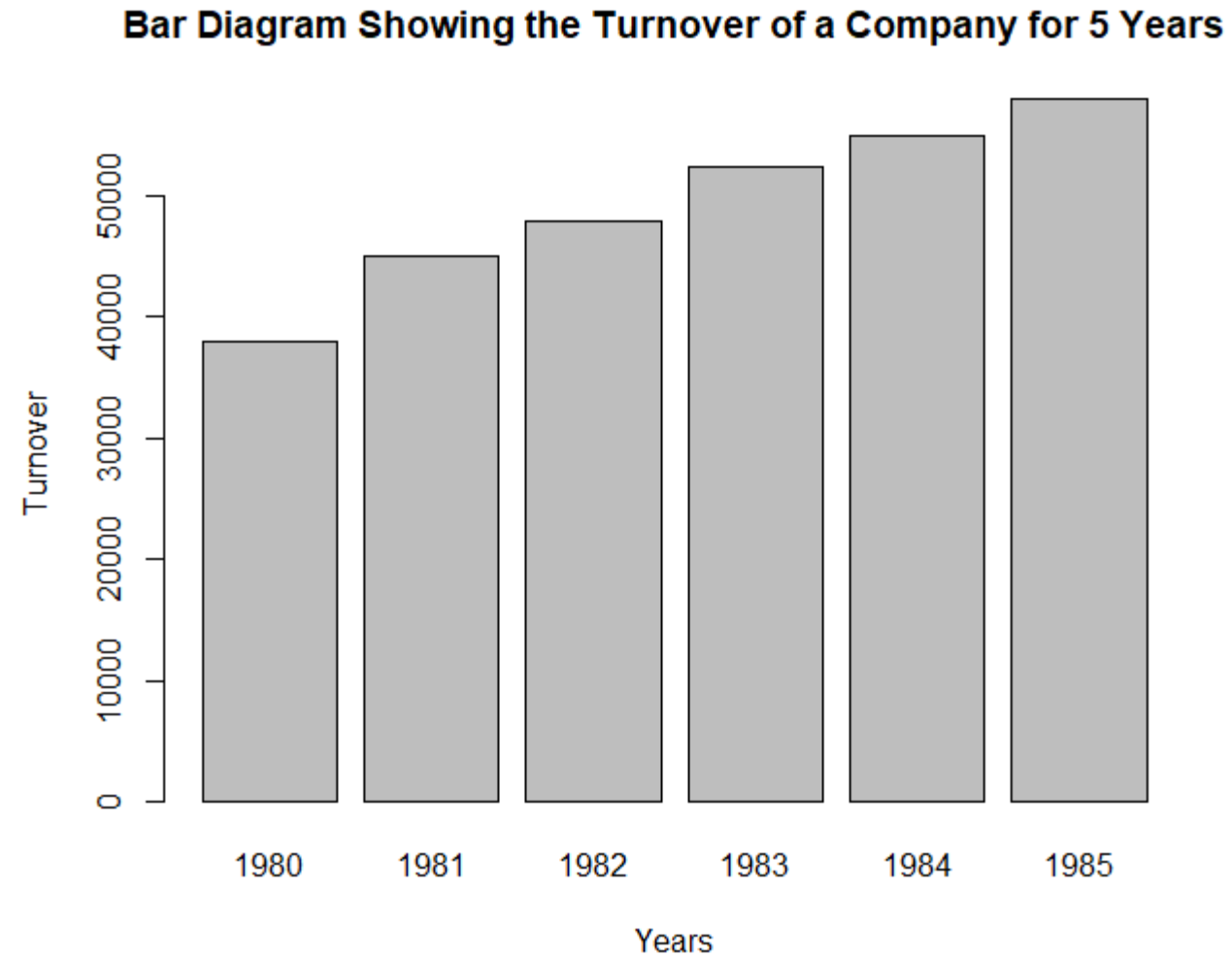
Example:

Draw a simple bar chart to represent the turnover of a company for 6 years.

Years	1980	1981	1982	1983	1984	1985
Turnover (In Rupees)	38,000	45,000	48,000	52,500	55,000	58,000

Data Presentation

- Simple Bar Chart



Data Presentation

- Pie Diagram

A pie diagram, also known as sector diagram, is a graphic device consisting of a circle divided into sectors whose areas are proportional to the various parts in which the whole quantity is divided.

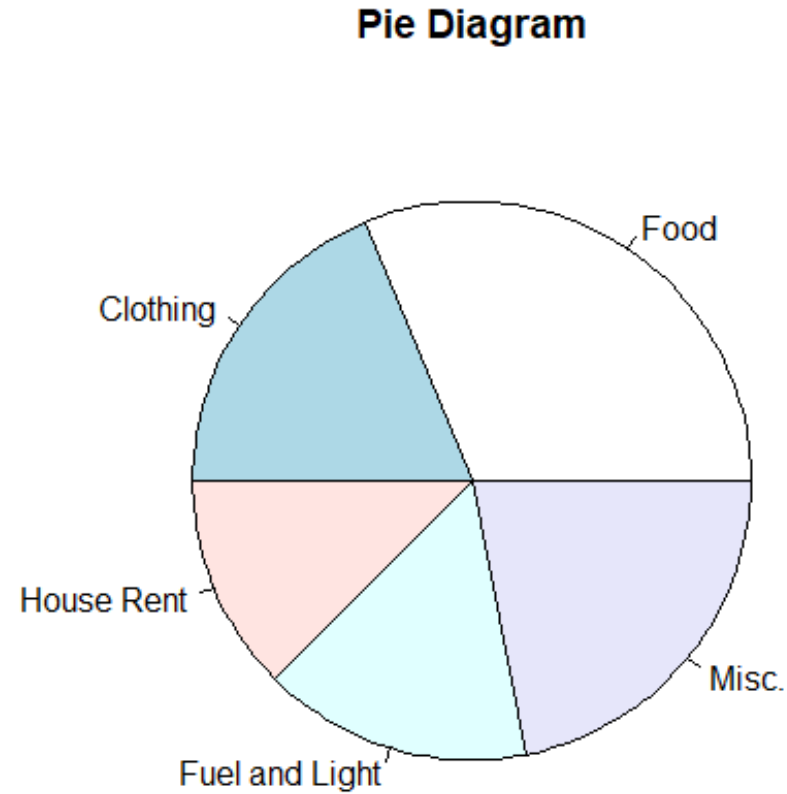
$$\text{Angle of Sector} = \frac{\text{Component Part}}{\text{Whole Quantity}} \times 360^{\circ}$$

Example: Represent the total expenditure and expenditure on various items of a family by pie diagram:

Items	Food	Clothing	House Rent	Fuel and Light	Misc.
Expenditure (In Rs.)	50	30	20	15	35

Data Presentation

- Pie Diagram



Data Presentation

- Graph of Time Series-Histogram

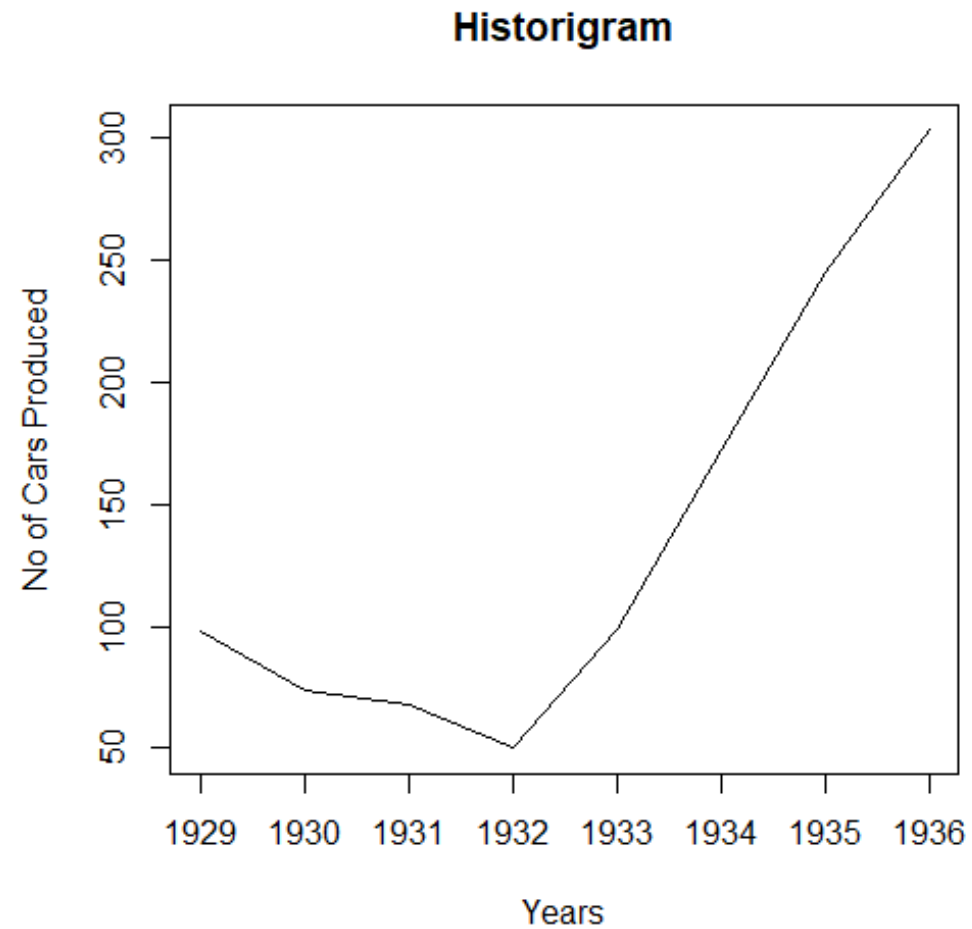
A curve showing the changes in the value of one or more items from one period of time to next is known as graph of time series.

The following table gives the number of cars produced in a city-C during the years 1929-1936. Draw the histogram of the series.

Years	1929	1930	1931	1932	1933	1934	1935	1936
No. of Cars	98	74	68	50	99	172	245	303

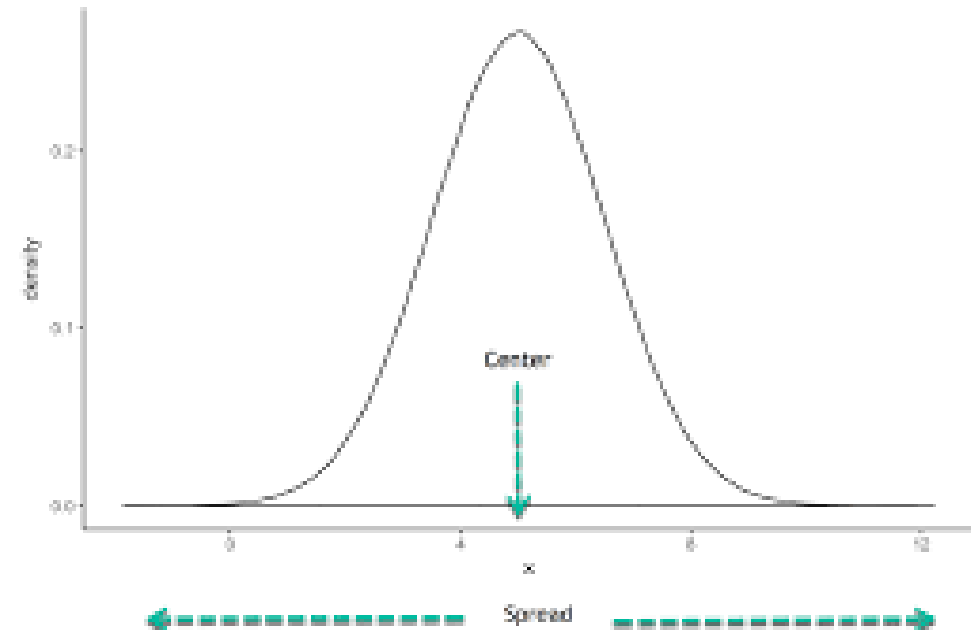
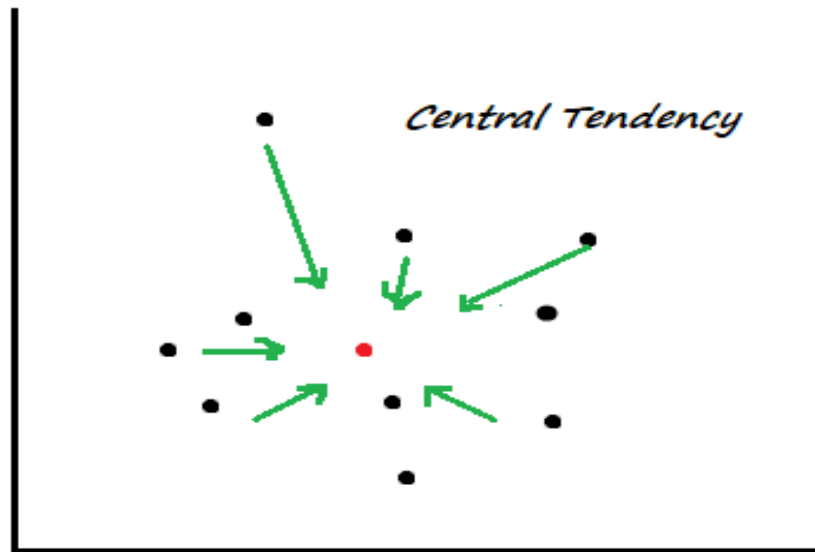
Data Presentation

- Graph of Time Series-Histogram



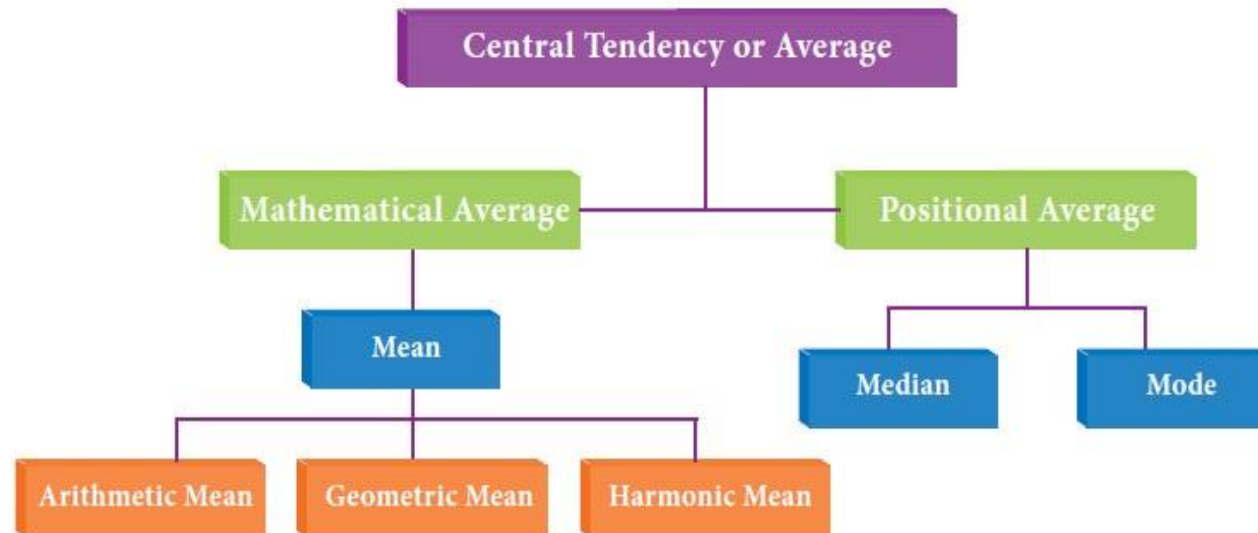
Measures of Central Tendency

Any measure of Central tendency is a single value that attempts to describe a set of data by identifying the central position within that set of data.



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Mathematical Averages

Mean (Arithmetic Mean for Ungroup Data):

The **mean** of a data set is the sum of the observations divided by the number of observations. The most used measure of center is the mean. When people speak of taking an average, they are most often referring to the mean.

Mean

For a variable x , the mean of the observations for a sample is called a **sample mean** and is denoted $\bar{x}$. Symbolically,

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$$

Data=Quiz Marks of 9 students in a class

DATA: { 16, 17, 10, 13, 20, 18, 13, 14, 18 }

$$\bar{x} = \frac{16 + 17 + 10 + 13 + 20 + 18 + 13 + 14 + 18}{9}$$

$$\bar{x} = \frac{139}{9}$$

$$\bar{x} = 15.444 \text{ (rounded to three decimal places)}$$

Properties of Arithmetic Mean

i) Sum of the deviations of observations from their mean is always zero.

$$\sum_{i=1}^n (X_i - \bar{X}) = 0$$

ii) Sum of squared deviations of values from their mean is always minimum
(comparative to any other average, A).

$$\sum_{i=1}^n (X_i - \bar{X})^2 < \sum_{i=1}^n (X_i - A)^2$$

iii) Change of origin and scale affect the mean.

If Y is a new transformed variable, $Y = a + bX$ then the mean of Y $\rightarrow \bar{Y} = a + b\bar{X}$

iv) The combined mean for more than one mean can be presented as:

$$\bar{X}_c = \frac{n_1\bar{X}_1 + n_2\bar{X}_2 + \dots + n_k\bar{X}_k}{n_1 + n_2 + \dots + n_k}$$

Properties of Arithmetic Mean

- i) Sum of the deviations of observations from their mean is always zero.

$$\sum_{i=1}^n (X_i - \bar{X}) = 0$$

- ii) Sum of squared deviations of values from their mean is always minimum

(comparative to any other average, A).

$$\sum_{i=1}^n (X_i - \bar{X})^2 < \sum_{i=1}^n (X_i - A)^2$$

Suppose X=Quiz marks of students in a class

X	X-Mean	Squared Errors
16	0.556	0.309
17	1.556	2.421
10	-5.444	29.637
13	-2.444	5.973
20	4.556	20.757
18	2.556	6.533
13	-2.444	5.973
14	-1.444	2.085
18	2.556	6.533
Mean=15.444	Sum of Errors = 0.004	Sum of Squared Errors = 80.222

X	X-A, A=14	Squared Errors
16	2	4
17	3	9
10	-4	16
13	-1	1
20	6	36
18	4	16
13	-1	1
14	0	0
18	4	16
Mean=15.444	Sum of Errors=13	Sum of Squared Errors = 99

Properties of Arithmetic Mean

iii) Change of origin and scale affects the mean.

$$Y = a + bX \quad \rightarrow \quad \bar{Y} = a + b\bar{X}$$

iv) The combined mean for more than one mean can be presented as:

$$\bar{X}_c = \frac{n_1\bar{X}_1 + n_2\bar{X}_2 + \dots + n_k\bar{X}_k}{n_1 + n_2 + \dots + n_k}$$

X	Y=2+3X
16	50
17	53
10	32
13	41
20	62
18	56
13	41
14	44
18	56
Mean=15.444 Mean(Y) = 2+3(15.444)	
Mean(Y) = 48.333	

X1	X2	X3
16	15	12
17	16	16
10	14	20
13	13	18
20	12	12
18	13	12
13	18	15
14	18	16
18	20	20
Mean(X1) = 15.444 Mean(X2) = 15.444 Mean(X3) = 15.667		
Combined Mean = 15.518		

Example: Combined Mean

- The mean heights and the number of students in three sections of a statistics class are given below:

Section	Number of Students	Mean Heights
A	40	62''
B	37	58''
C	43	61''

- Find the overall mean height of 120 students.

Mathematical Averages

Mean (Arithmetic Mean for Grouped Data)

When the data is presented in terms of classes and their frequencies, slightly different formula is applicable to find out the average.

Frequency distribution of car battery life:

Class Interval	Class Mid Point (X)	Frequency (f)	f*X
1.5 - 1.9	1.7	2	3.4
2.0 - 2.4	2.2	1	2.2
2.5 - 2.9	2.7	4	10.8
3.0 - 3.4	3.2	15	48
3.5 - 3.9	3.3	10	33
4.0 - 4.4	4.2	5	21
4.5 - 4.9	4.7	3	14.1
		Sum(f) = 40	Sum(f*X)=132.5

Average Battery Life:

$$Mean = \frac{\sum fX}{\sum f} = \frac{132.5}{40} = 3.3125$$

Example: Mean of the Grouped Data

- The following distribution shows Kilowatt-Hours of electricity used in one month by 75 residential customers in a certain locality of a city G.

Consumption	No. of Customers
5-24	4
25-44	6
45-64	14
65-84	22
85-104	14
105-124	5
125-144	7
145-164	3

(i) Estimate the mean consumption of electricity

(ii) Show that $\sum_{i=1}^n f(X_i - \bar{X}) = 0$

Mathematical Averages

Minerals are transferred from the fungus to the trees and sugars from the trees to the fungus. Two samples of 10 northern red oak seedlings were planted in a greenhouse, one containing seedlings treated with nitrogen the other containing seedlings with no nitrogen.

Trimmed Mean

A trimmed mean is computed by “trimming away” a certain percent of both the largest and the smallest set of values. For example, the 10% trimmed mean is found by eliminating the largest 10% and smallest 10% and computing the average of the remaining values.

Table 1.1: Data Set for Example 1.2

No Nitrogen	Nitrogen
0.32	0.26
0.53	0.43
0.28	0.47
0.37	0.49
0.47	0.52
0.43	0.75
0.36	0.79
0.42	0.86
0.38	0.62
0.43	0.46

10% trimmed mean weight for the with-nitrogen group:

$$\bar{X}_{tr(10)} = \frac{0.32 + 0.37 + 0.47 + 0.43 + 0.36 + 0.42 + 0.38 + 0.43}{8}$$

10% trimmed mean weight for the without-nitrogen group:

$$\bar{X}_{tr(10)} = \frac{0.47 + 0.43 + 0.49 + 0.52 + 0.75 + 0.79 + 0.62 + 0.46}{8}$$

Simple Bar Chart in R

- `Years<-c("1980","1981","1982","1983","1984","1985")`
- `TurnOver<-c(38000,45000,48000,52500,55000,58000)`
- `names(TurnOver)<-Years`
- `barplot(TurnOver,xlab="Years",ylab="Turnover",main="Bar Diagram Showing the Turnover of a Company for 5 Years")`

Pie Chart in R

- `Items<-c("Food","Clothing", "House Rent","Fuel and Light", "Misc.")`
- `Exp<-c(50,30,20,25,35)`
- `names(Exp)<-Items`
- `pie(Exp, main="Pie Diagram")`

Historigram in R

- `Years<-c(1929,1930,1931,1932,1933,1934,1935,1936)`
- `Cars<-c(98,74,68,50,99,172,245,303)`
- `plot(Years,Cars, type="l",main="Historigram",xlab="Years",ylab="No of Cars Produced")`

Frequency Distribution in R

Data used in the above example:

- `X <- c(2.2,4.1,3.5, 4.5,3.2,3.7,3,2.6,3.4,1.6,3.1,3.3,3.8,3.1,4.7,3.7,2.5,4.3,3.4, 3.6,2.9,3.3,3.9,3.1,3.3,3.1,3.7,4.4,3.2,4.1,1.9,3.4,4.7,3.8,3.2,2.6,3.9,3,4.2,3.5)`
- `min(X)`
- `max(X)`
- `bins <- seq(min(X),max(X), by=0.5)`
- `Life <- cut(X, bins)`
- `table(Life)` `## for horizontal output`
- `transform(table(Life))` `## for vertical output`

Creating Histogram in R

Histograms

The generic function “hist” computes a histogram of the given data values. It can be used as:

hist(x, ...)

Example 1:

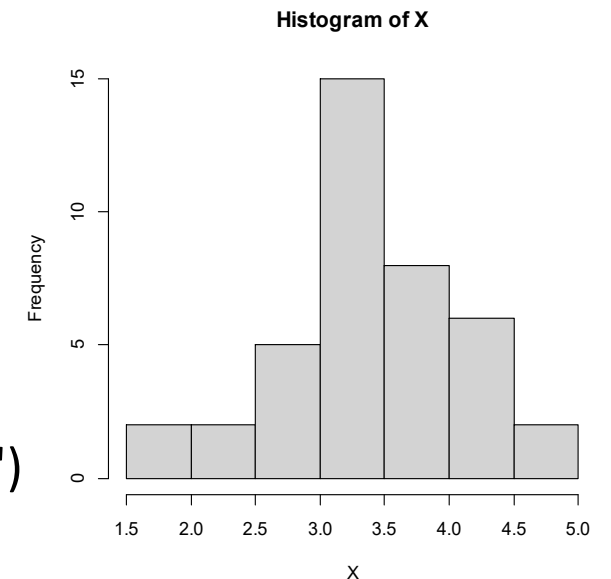
```
X<-c(2.2,4.1,3.5,4.5,3.2,3.7,3,2.6,3.4,1.6,3.1,3.3,3.8,3.1,4.7,3.7,2.5,4.3,3.4,3.6,2.9,3.3,3.9,3.1,3.3,3.1,  
3.7,4.4,3.2, 4.1,1.9,3.4,4.7,3.8,3.2,2.6,3.9,3,4.2,3.5)
```

```
hist(X)
```

Example 2:

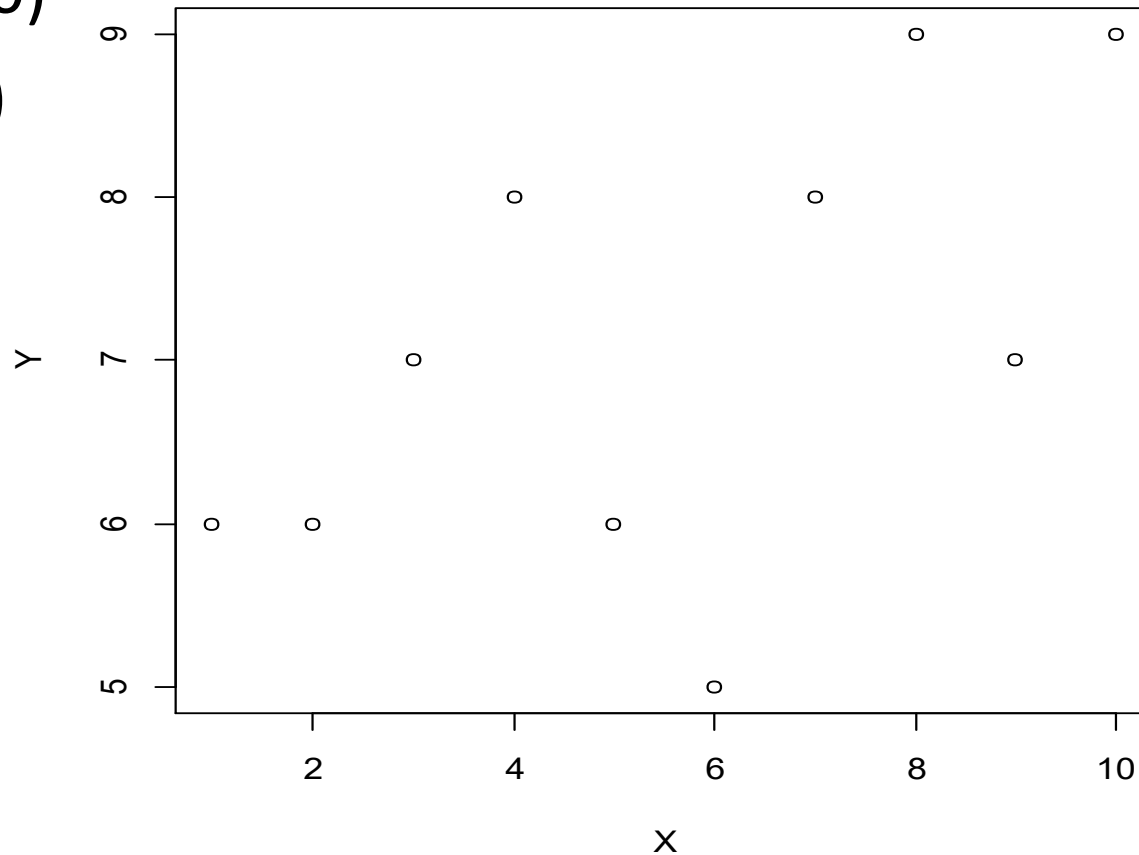
```
X <- rnorm(200, 10, 1)
```

```
hist(X, main = "Histogram of Normal Random Variable", breaks = "Sturges")
```



Scatter Plot in R

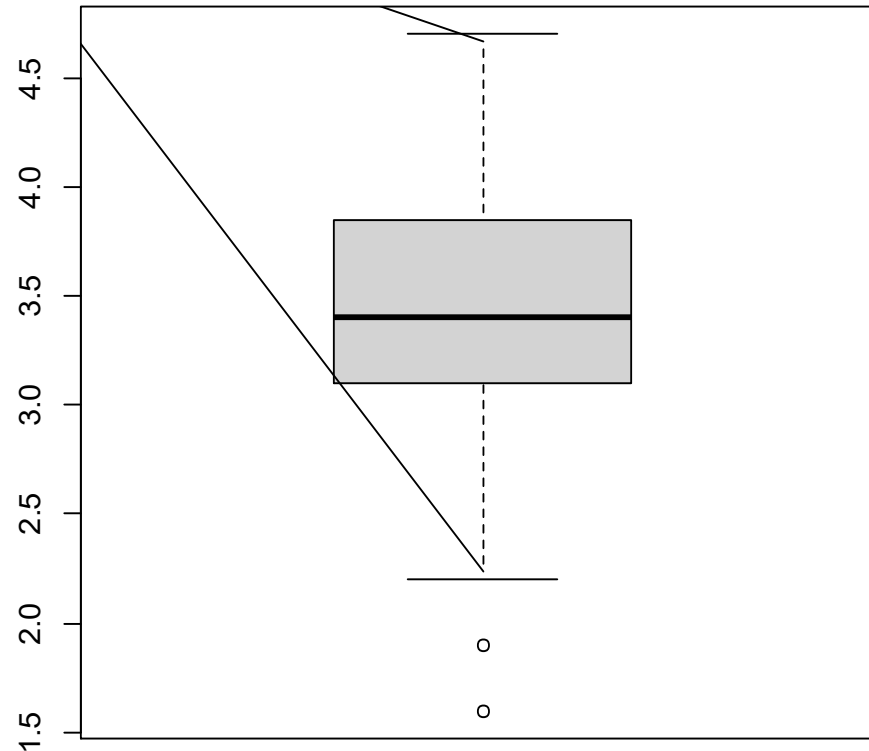
- `X <- c(1,2,3,4,5,6,7,8,9,10)`
- `Y <- c(6,6,7,8,6,5,8,9,7,9)`
- `plot(X,Y)`



Creating Box-Plot in R

```
X<-c(2.2,4.1,3.5,4.5,3.2,3.7,3,2.6,3.4,1.6,3.1,3.3,3.8,3.1,4.7,3.7,2.5,4.3,  
3.4,3.6,2.9,3.3,3.9,3.1,3.3,3.1, 3.7,4.4,3.2, 4.1,1.9,3.4,4.7, 3.8,3.2,2.6,  
3.9,3,4.2,3.5)
```

```
boxplot(X)
```



Creating Box-Plot in R

```
X<-c(2.2,4.1,3.5,4.5,3.2,3.7,3,2.6,3.4,1.6,3.1,3.3,3.8,3.1,4.7,3.7,2.5,4.3,  
3.4,3.6,2.9,3.3,3.9,3.1,3.3,3.1, 3.7,4.4,3.2, 4.1,1.9,3.4,4.7, 3.8,3.2,2.6,  
3.9,3,4.2,3.5)
```

stem(X)

```
> X<-c(2.2,4.1,3.5,4.5,3.2,3.7,3,2.6,3.4,1.6,3.1,3.3,3.8,3.1,4.7,3.7,2.5,4.3, 3$  
> stem(X)
```

```
The decimal point is at the |
```

```
1 | 69
```

```
2 | 2
```

```
2 | 5669
```

```
3 | 001111222333444
```

```
3 | 5567778899
```

```
4 | 11234
```

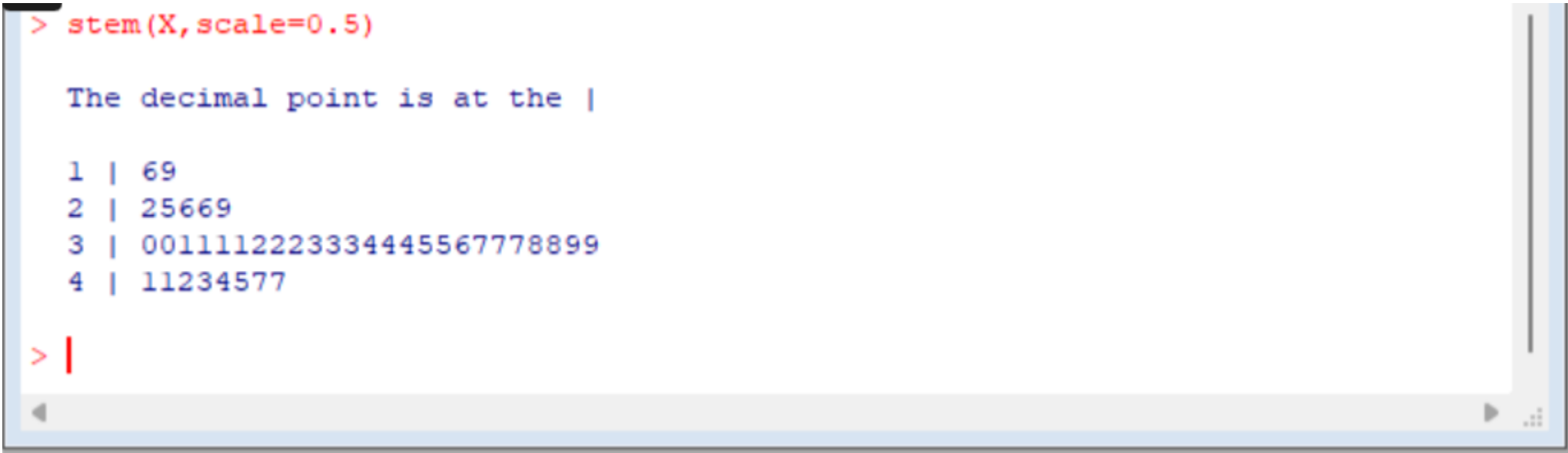
```
4 | 577
```

```
> |
```

Creating Box-Plot in R

```
X<-c(2.2,4.1,3.5,4.5,3.2,3.7,3,2.6,3.4,1.6,3.1,3.3,3.8,3.1,4.7,3.7,2.5,4.3,  
3.4,3.6,2.9,3.3,3.9,3.1,3.3,3.1, 3.7,4.4,3.2, 4.1,1.9,3.4,4.7, 3.8,3.2,2.6,  
3.9,3,4.2,3.5)
```

```
stem(X)
```



```
> stem(X,scale=0.5)  
  
The decimal point is at the |  
  
1 | 69  
2 | 25669  
3 | 0011112223334445567778899  
4 | 11234577  
  
> |
```